06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

1.0.1 The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel's spend over time (usually weekly), used to attribute sales to channels and set budgets. It's the "top-down" alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn't only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an "effective" exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM fixes.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

1.0.2 Why "causal" MMM

A plain MMM regresses sales on all channels and can be fooled by a **confounder** like seasonality, which lifts sales *and* drives brand-search spend (people search more in peak season). A **causal MMM** takes a **DAG** of the market (which channels cause what) and uses the backdoor logic from notebook 05 to control for genuine confounders while *not* adjusting away legitimate mediated effects — so it separates real incremental contribution from spurious credit.

Two honest themes run through this notebook: (1) the **confounding correction is robust and decisive** — it gets the *ranking* and the go/no-go right; (2) an MMM's **absolute** contribution levels are *not* reliable from observational data alone — so you **calibrate them with a geo experiment** (exactly Anchor B, notebook 07). Saying this out loud is the mark of an honest MMM.

> **On real data.** Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta's *Robyn* or `pymc-marketing`'s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

7-step contract.

1.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave, so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

```
TRUE incremental sales - TV 3,536   brand_search 333 (small)
```

```
   date_week         tv  brand_search  seasonality       sales
0 2023-01-02  42.870187     21.749830     0.000000  130.875186
1 2023-01-09  53.049712     18.348364     0.120537  136.412935
2 2023-01-16  27.623875     25.155597     0.239316  135.190674
3 2023-01-23  56.329747     32.172097     0.354605  152.617219
4 2023-01-30  43.339248     35.793793     0.464723  156.933276
```

1.2 3 · Identify — the marketing DAG picks controls vs mediators

DAG: `seasonality -> {sales, brand_search}`, `tv -> sales`, `brand_search -> sales`. The backdoor criterion says **seasonality is a confounder -> control for it**; omitting it is exactly the last-click mistake that over-credits brand search. The estimand for channel c is the interventional contribution $\mathbb{E}[\text{sales} \mid do(x_c)]$ − baseline; ROI = contribution ÷ spend, each with a posterior. We hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

```
Backdoor adjustment set the causal layer will use: ['seasonality']
```

1.3 4 · Estimate — fit the causal MMM

A caution up front, and an honest one: **MMMs are notoriously miscalibrated in *absolute* terms** — with channel spend correlated with seasonal demand, a short weekly series can't fully separate "channel drove sales" from "season drove both", so fitted contributions run *higher* than truth. That's why practitioners **calibrate MMMs with geo experiments** (nb 07). What the MMM *does* get right, and what drives budget, is the **ranking** and each channel's **ROI verdict**.

```
Output()
```

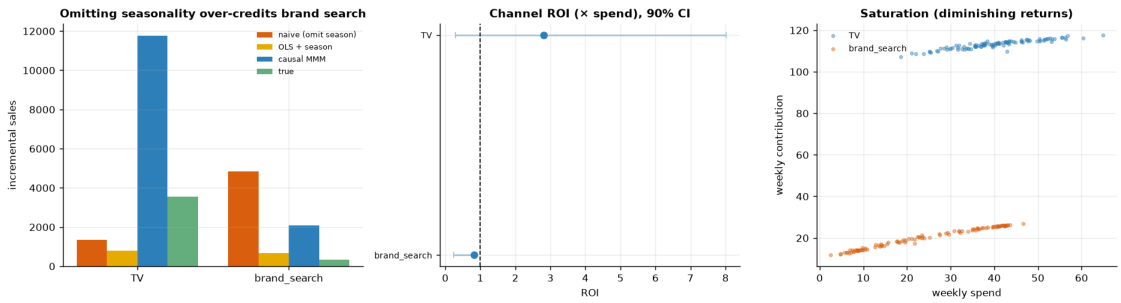
```
Causal MMM contribution - TV 11,775   brand_search 2,083
(true: TV 3,536, brand_search 333 - absolute levels run high; see note.)
```

1.4 5 · Validate — confounding correction, ROI with uncertainty, and saturation

Three views:

1. **The confounding correction** (robust): naive attribution *omitting* seasonality over-credits brand_search; adjusting for it (what the DAG prescribes) collapses that credit toward the truth.
2. **ROI with uncertainty** — a forest of each channel's posterior ROI vs the break-even line; this is the decision-relevant quantity, and it's far more stable than the absolute contribution.
3. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that determines *marginal* ROI and hence where the next euro should go).

```
brand_search credit - naive 4,843 | adjusted 679 | true 333
ROI - TV 2.81x [90% 0.29,8.02]  ·  brand_search 0.82x [90% 0.24,0.89]
```



How to read this. *Left* is the headline confounding correction, and it's robust: the **naive** attribution (orange) that ignores seasonality massively over-credits brand_search, while **adjusting for seasonality** (gold) collapses that credit toward the small truth. The causal MMM (blue) moves in the same direction but only *part-way* — it still lands above the OLS+season number, a live reminder that an MMM's *absolute* levels run high (which is exactly why we lean on the ranking, not the euros). *Middle* — the decision-relevant quantity, each channel's **ROI with a credible interval**: brand_search's interval sits **entirely below** break-even (1x), so its below-break-even verdict is solid; TV's posterior *mean* clears 1x (2.81x), but its 90% interval is **wide and dips below 1x**, so TV's *own* above-break-even verdict is genuinely uncertain. And the **ranking** is only *probably* right: draw-by-draw, TV's ROI beats brand_search's about **73%** of the time (the `P(TV ROI > brand_search ROI)` computed next) — more likely than not, but **short of the 0.8 confidence bar** we'd want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve's *slope*, not its average. The one honest caveat (Step 7): these ROI *levels* are approximate and should be anchored with a geo experiment — and, as the next step shows, even the *ranking* here isn't yet confident enough to act on.

1.5 6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental ROI**, not last-click credit. The point estimates rank TV above brand_search (brand_search mostly captures seasonal demand), so the *direction* of any reallocation is toward TV. But a direction is not yet a decision: we quantify the sales impact of shifting a slice of budget **and** the posterior probability that the move actually helps. If that probability clears our action bar we reallocate; if not — as happens here — the honest call is to **run a geo test first** (Anchor B, nb 07) and let it, not a noisy MMM, authorise the spend shift.

```
Shifting 15% of brand_search spend (€383) to TV -> projected +€764 sales
P(TV ROI > brand_search ROI) = 0.73 -> hold, test first
Caveat: this uses AVERAGE ROI; with saturation, marginal ROI falls as you pour
more into TV -
so reallocate in steps and re-estimate, and calibrate the absolute levels with a
geo experiment.
```

1.6 7 · Caveats

- **Absolute contributions need calibration.** Here they ran ~2–3x the truth because channel spend rides the seasonal wave the MMM can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — calibrate levels with a geo experiment (nb 07). This is the single most important thing to say about any MMM.
- **The DAG is the assumption.** If TV genuinely creates incremental search value, encode that edge or you'll under-credit it.
- **Adstock/saturation priors matter.** Carry-over (lambda) and diminishing-returns (alpha, kappa) are weakly identified from limited weekly data; use informative priors from experiments where you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When two channels move together, the data can't fully separate them; the intervals will (honestly) show it.